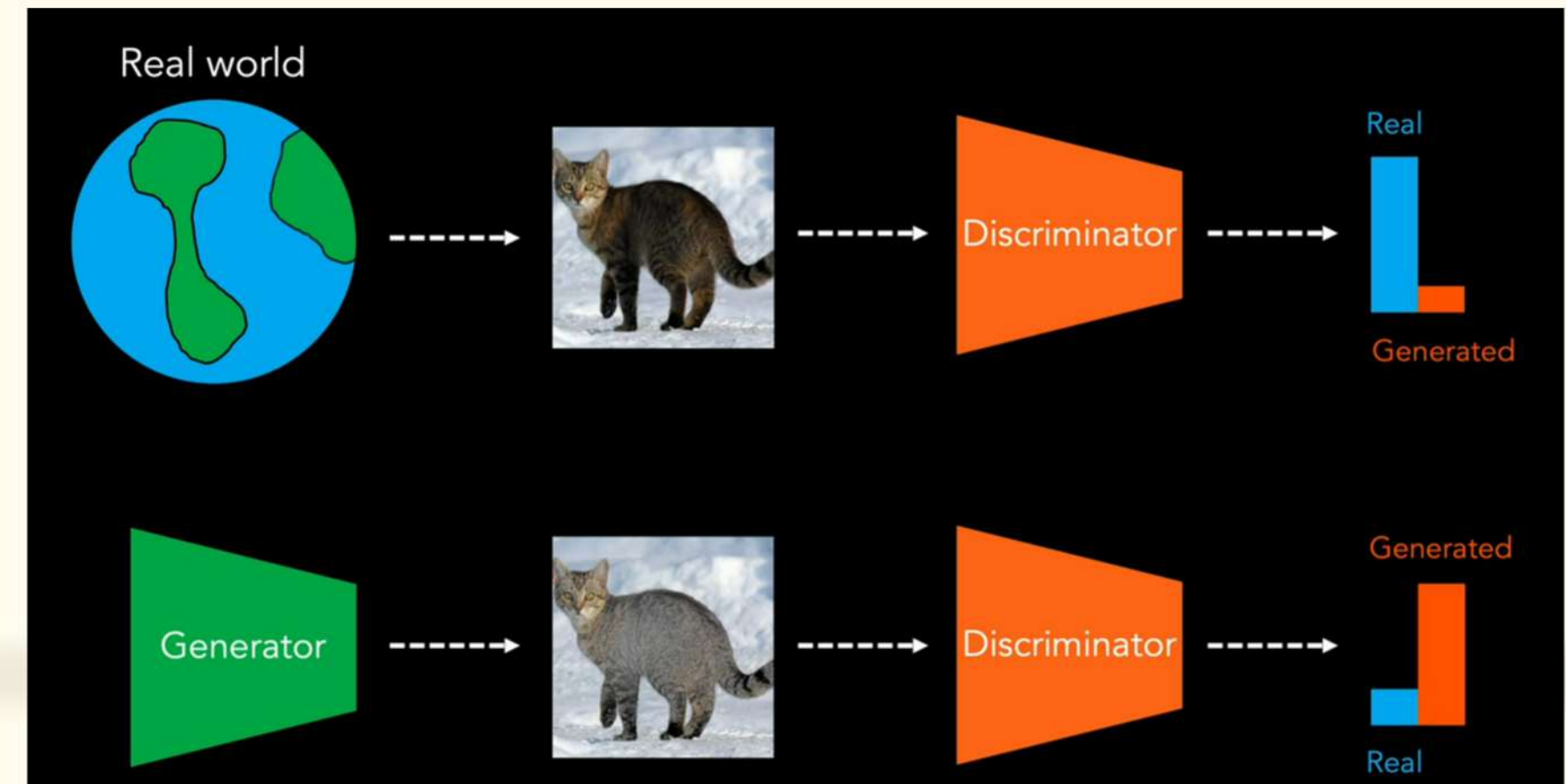
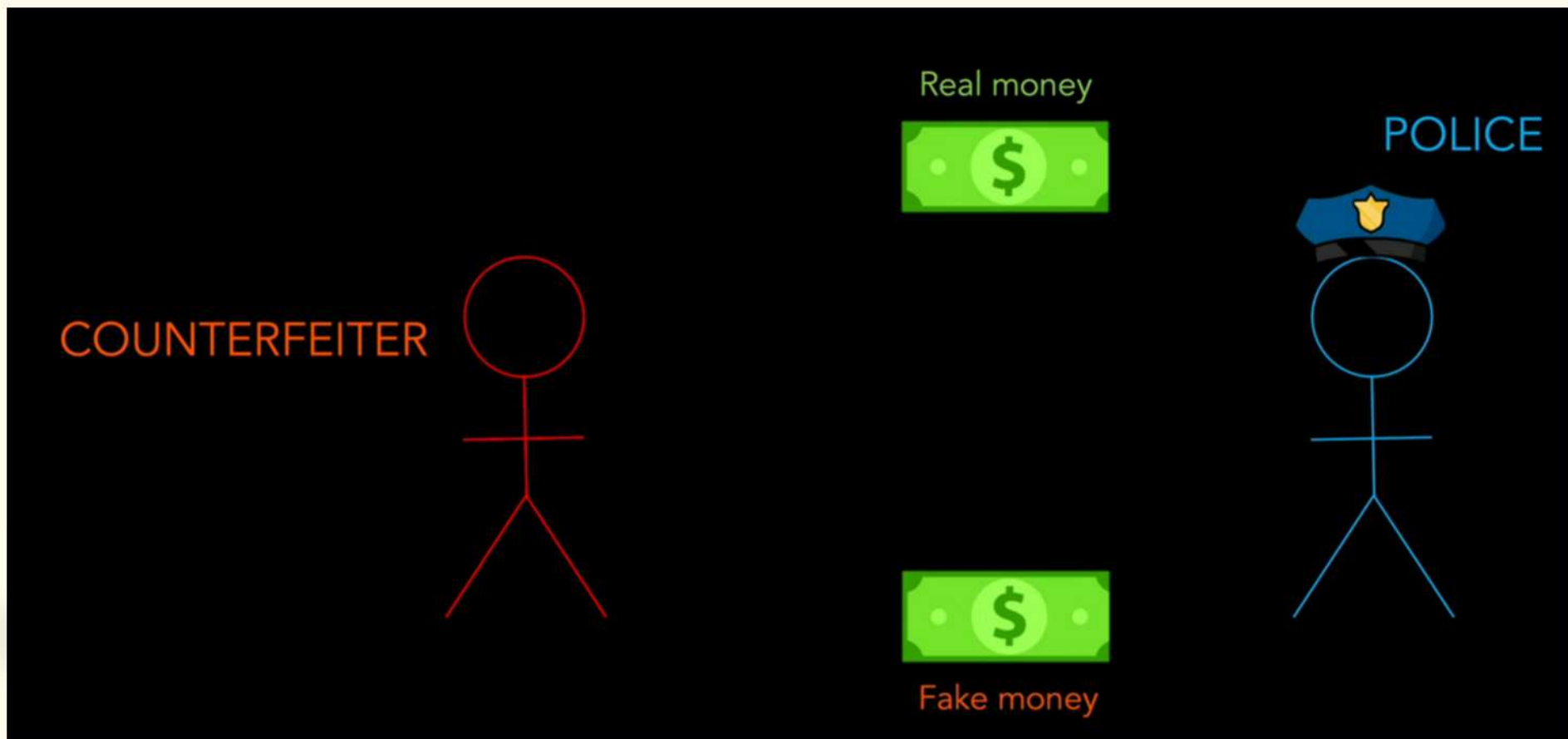


INTRODUCTION TO GAN



→ GAN are neural networks that are trained in an adversarial manner to generate data mimicking some distribution

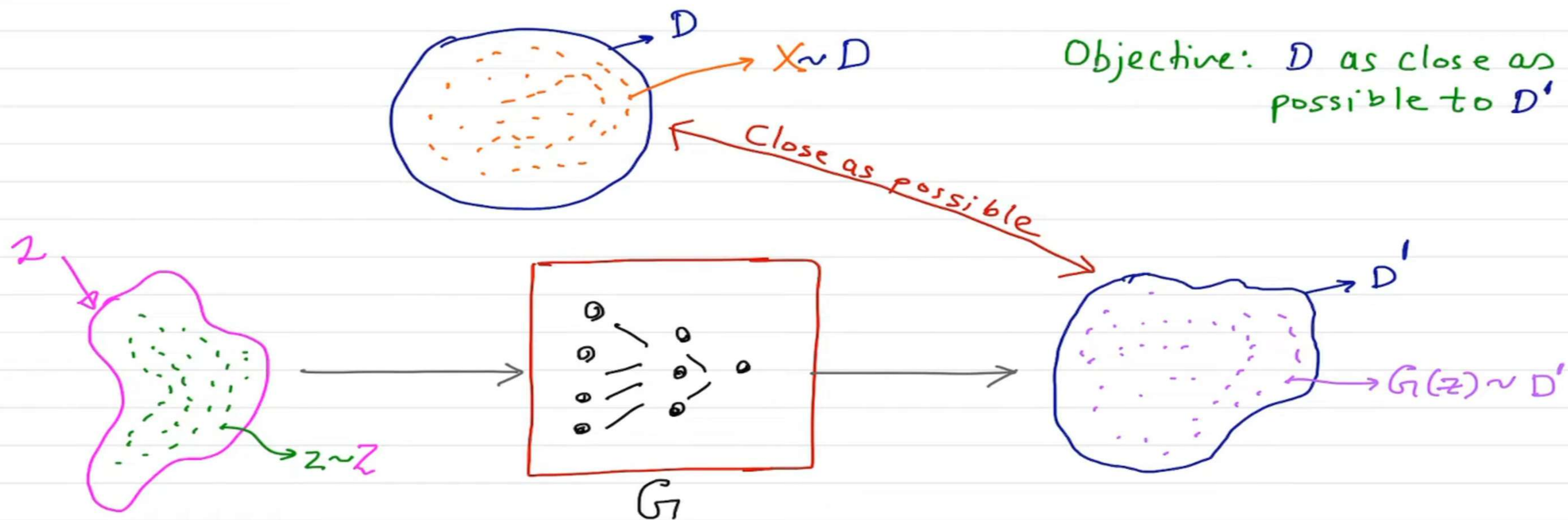
→ Two classes of models in machine learning

(a) Discriminative model: It is the one that discriminates between two different classes of data.

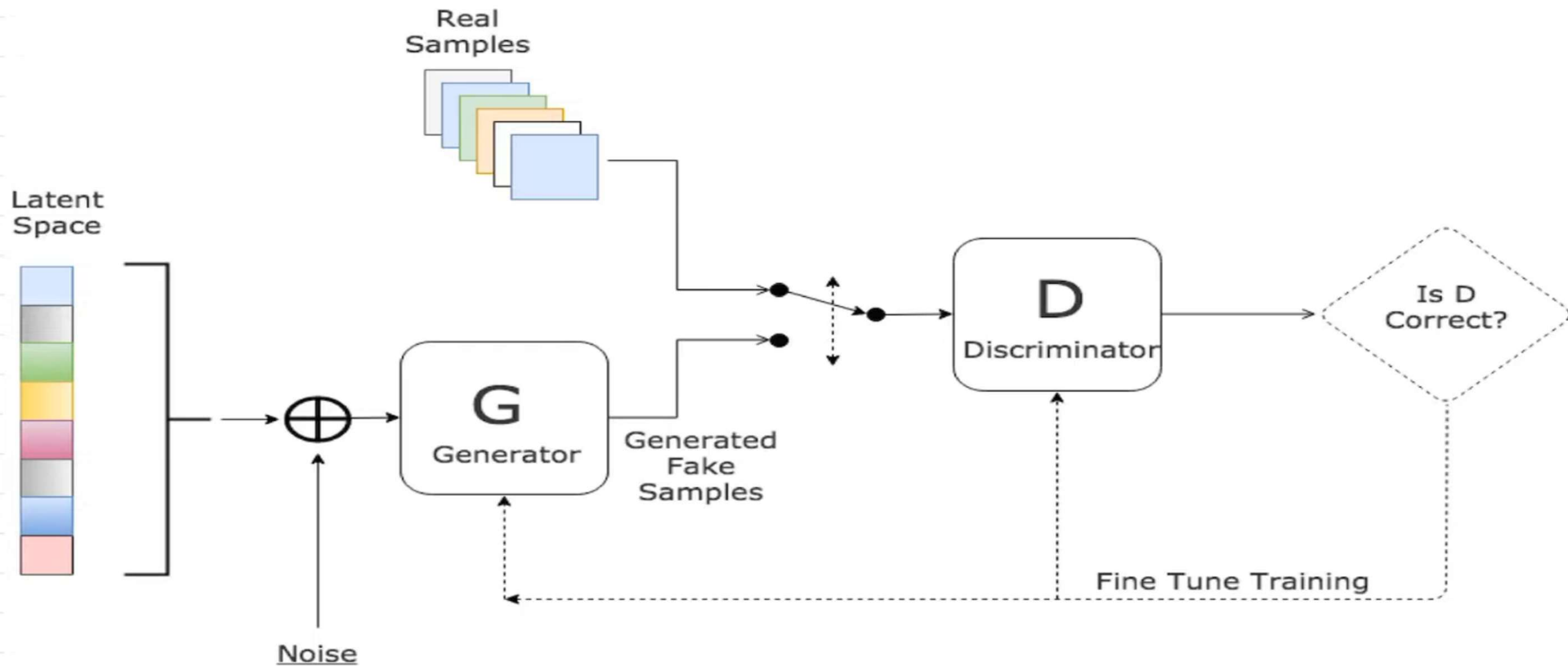
(b) Generative model: A generative model G to be trained on training data X sampled from some true distribution D is the one which, given some standard random distribution

Z produces a distribution D' which is close to D according to some closeness metric. Mathematically,

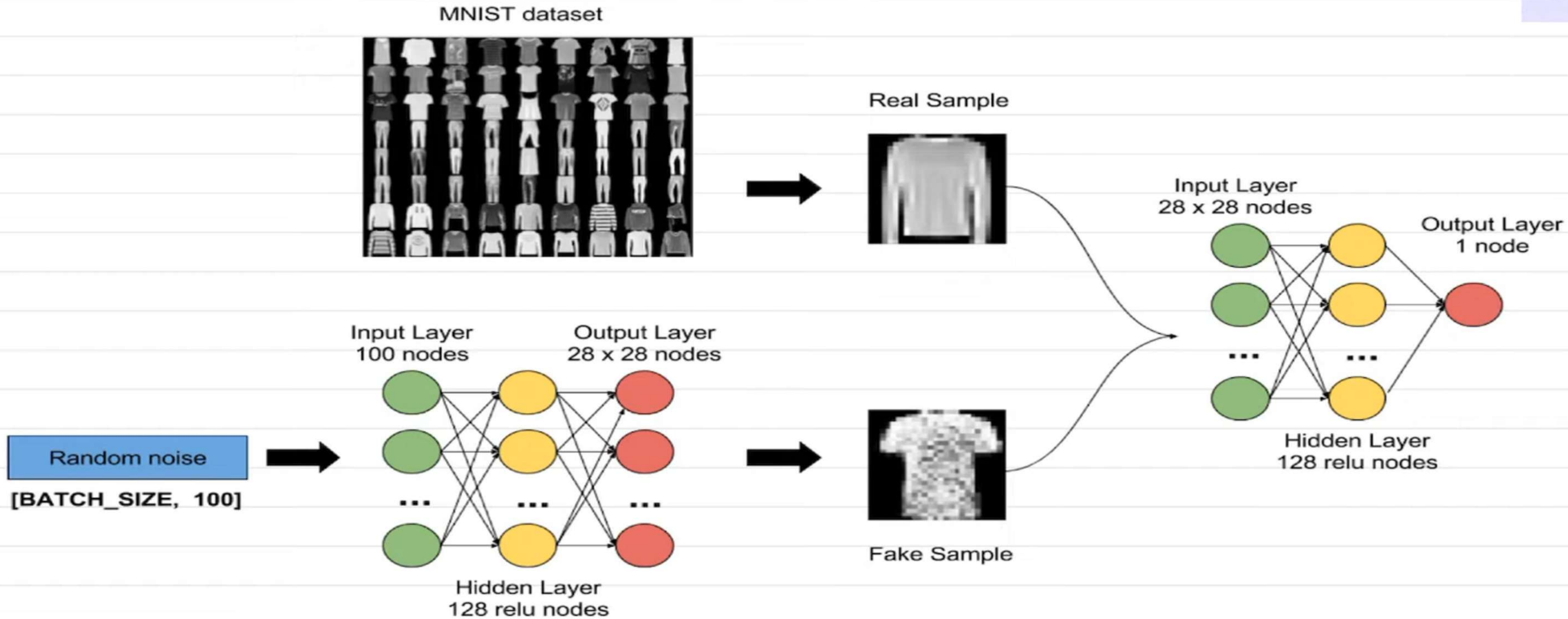
$z \sim Z$ maps to a sample $G(z) \sim D'$

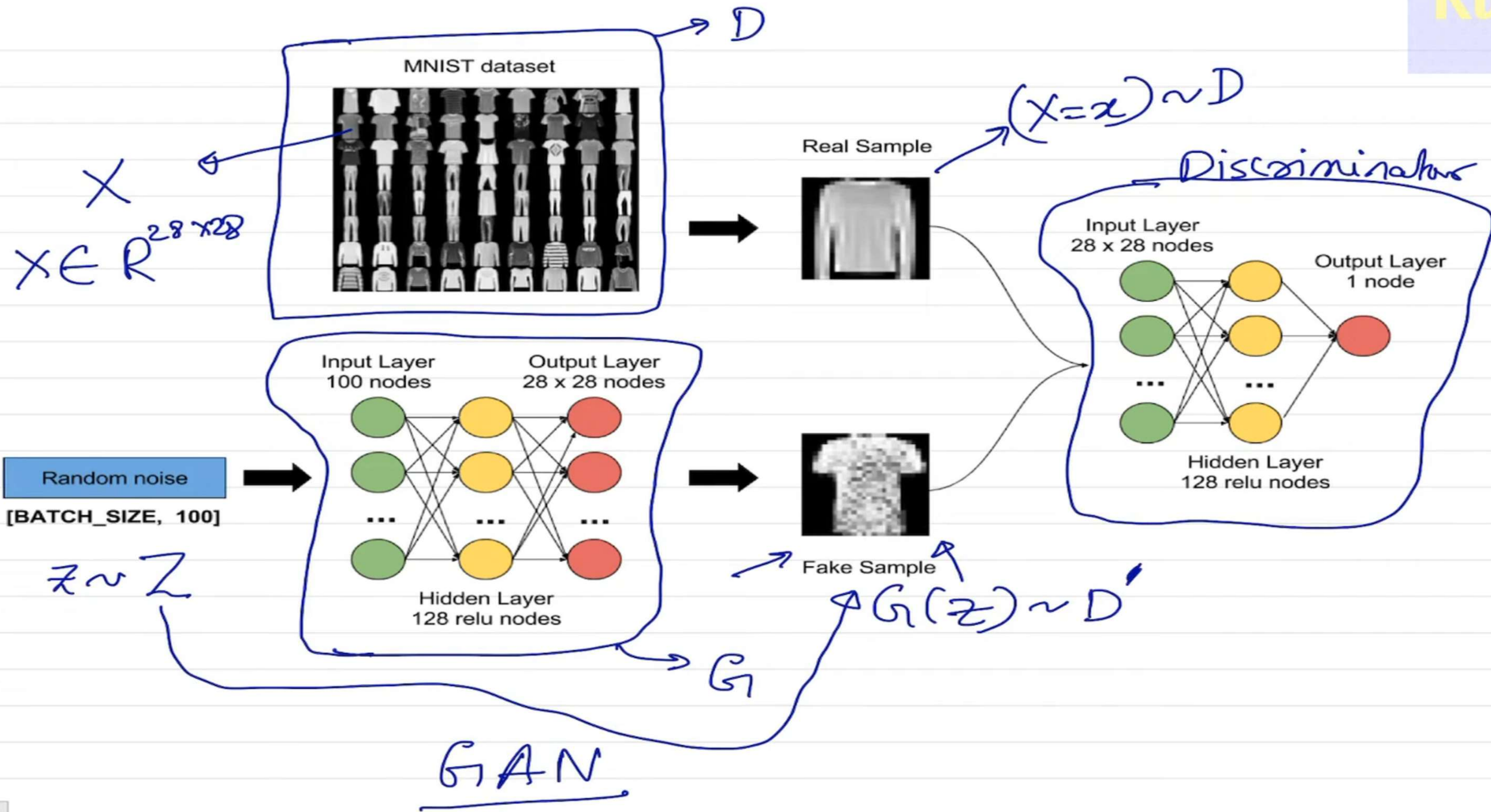


Generative Adversarial Network



EXAMPLE WITH MNIST DATASET

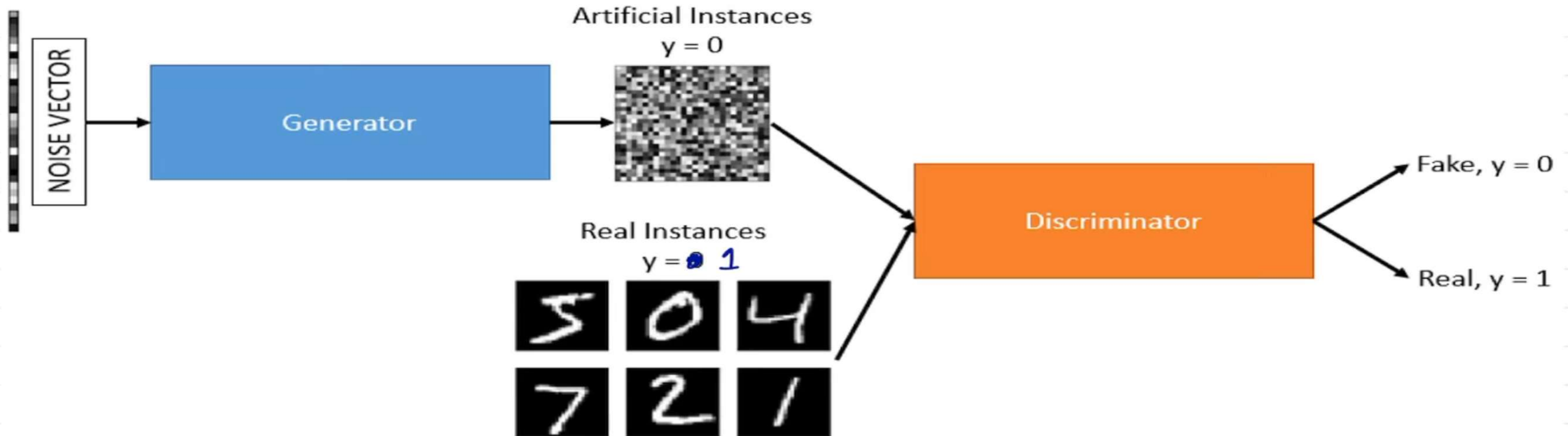




TRAINING A GAN

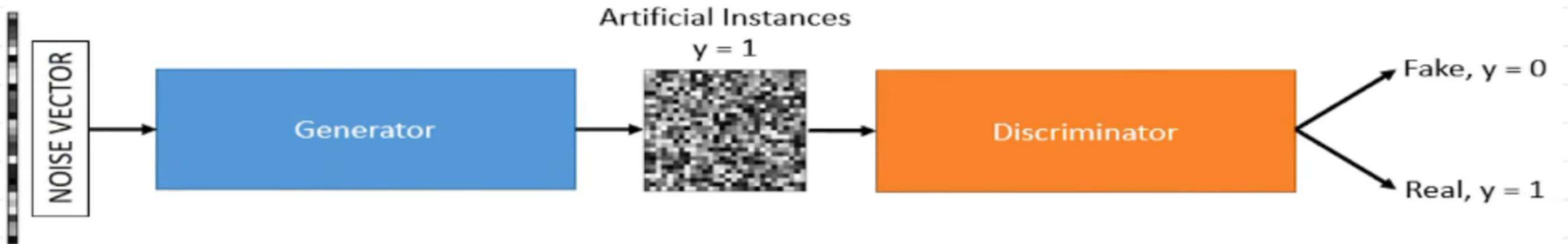
TRAINING DISCRIMINATOR AND GENERATOR SEPERATELY

TRAINING A DISCRIMINATOR



TRAINING A GENERATOR

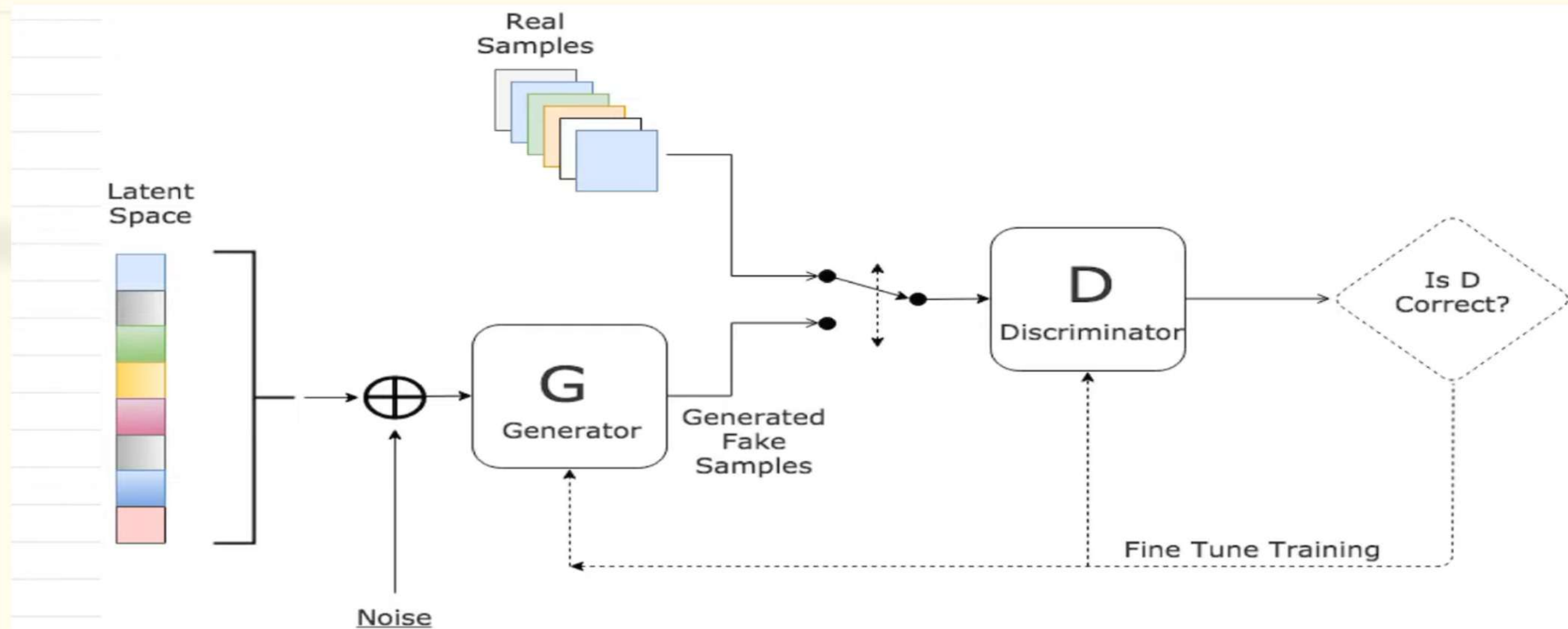
MAKE NO CHANGES IN DISCRIMINATOR



RECAP

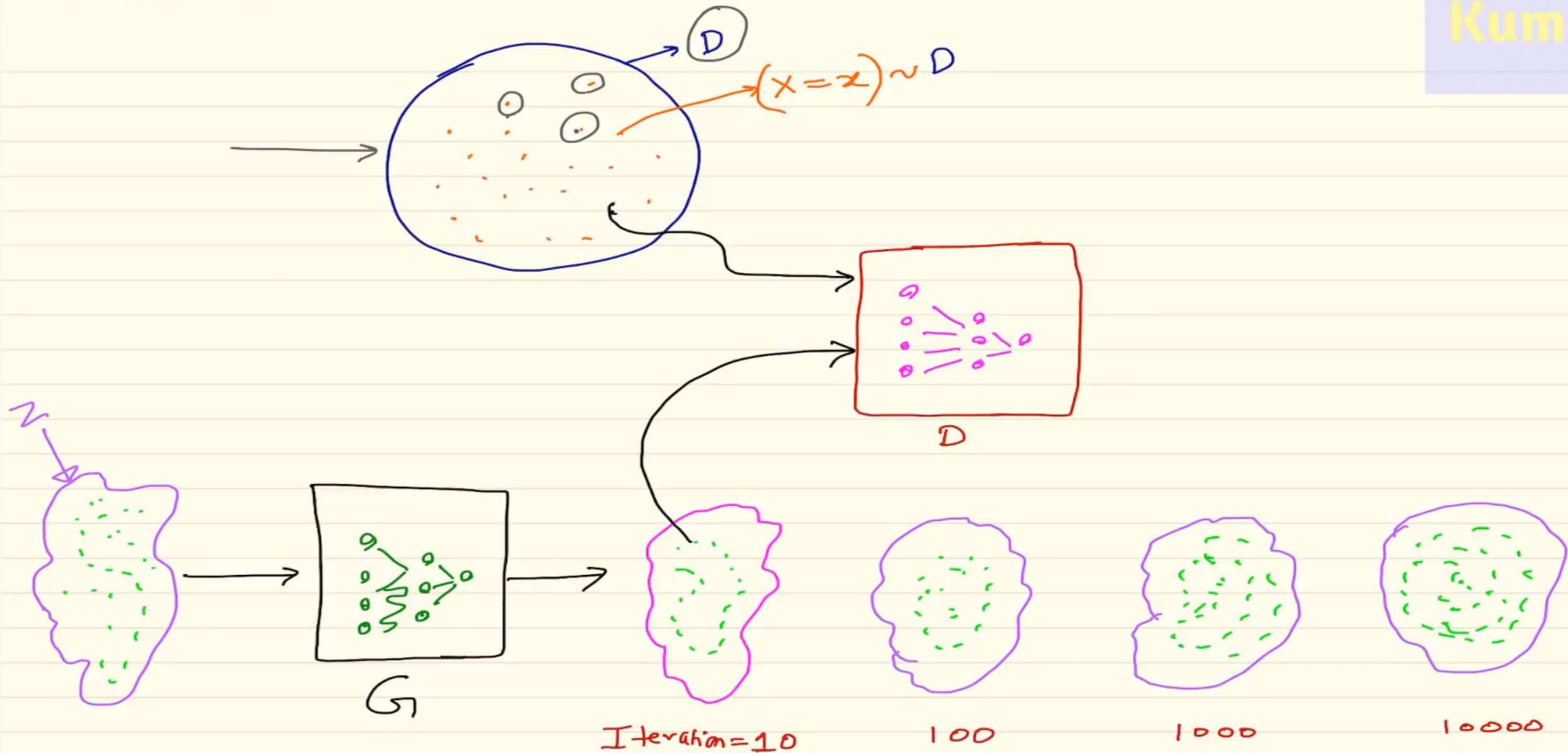
⊗ Discriminator: Role is to distinguish between actual data & fake data

⊗ Generator: Role is to create data in such a way so that it can fool the generator.

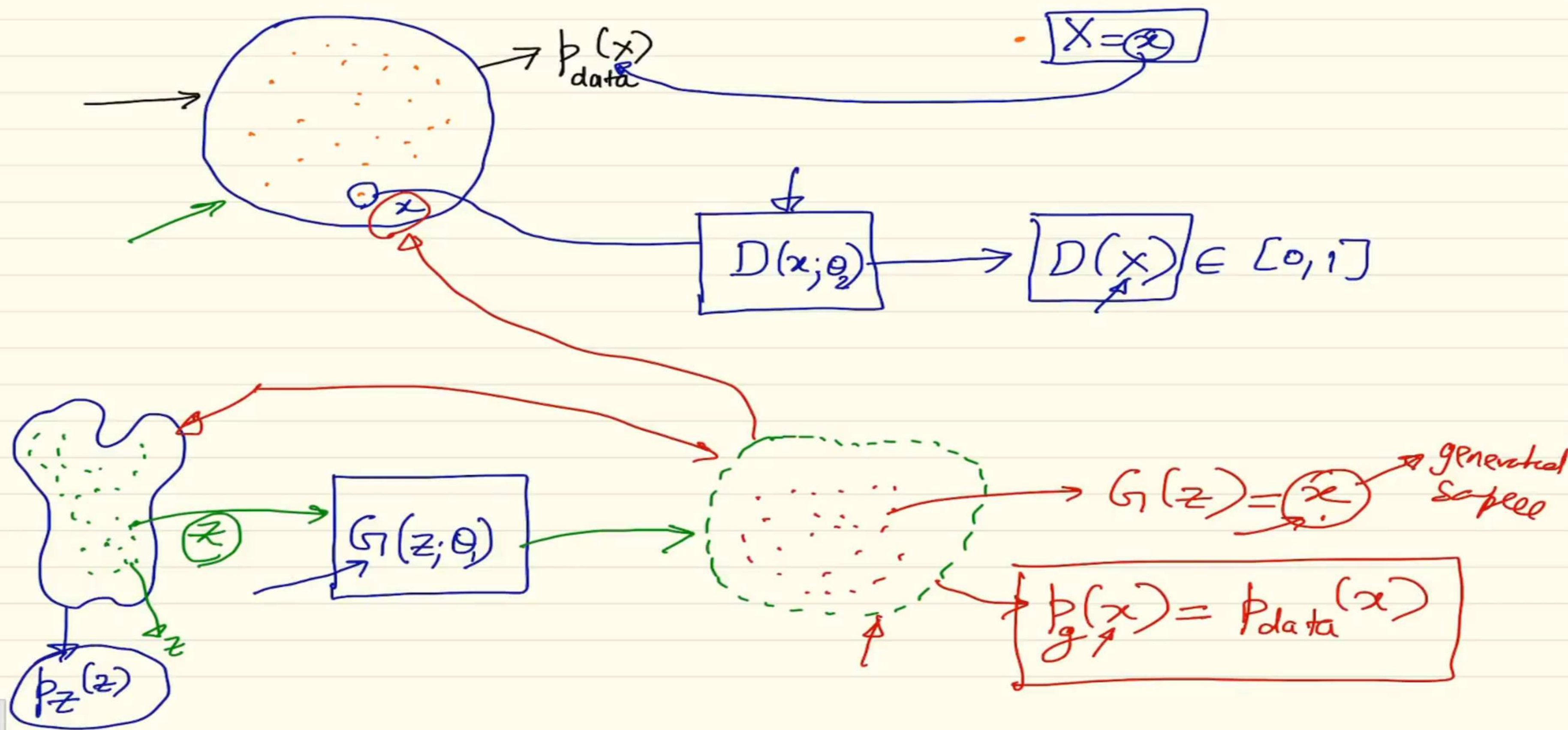


BACKPROPAGATION IDEA

Kume



Basic Conventions to understand Loss function



Loss function (Binary cross-entropy)

$$L(\hat{y}, y) = \left[y \log \hat{y} + (1-y) \log (1-\hat{y}) \right]$$

The label for the data coming from $p_{\text{data}}(x)$ is $y = 1$
& $\hat{y} = D(x)$ so putting this we obtain

$$L(D(x), 1) = \log(D(x)) - \textcircled{A}$$

& for data coming from generator the label is $y = 0$ & $\hat{y} = D(G(x))$
So in that case

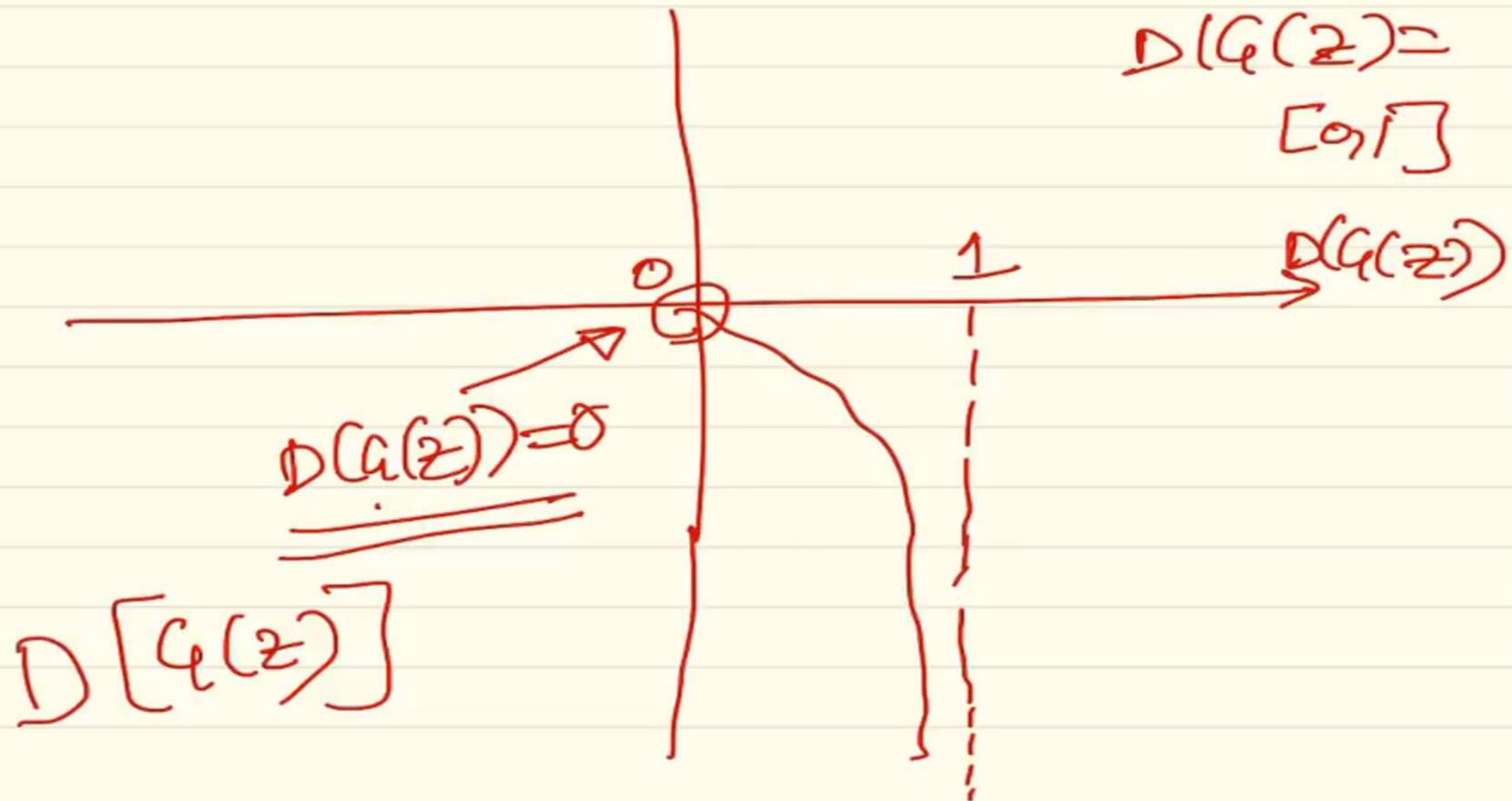
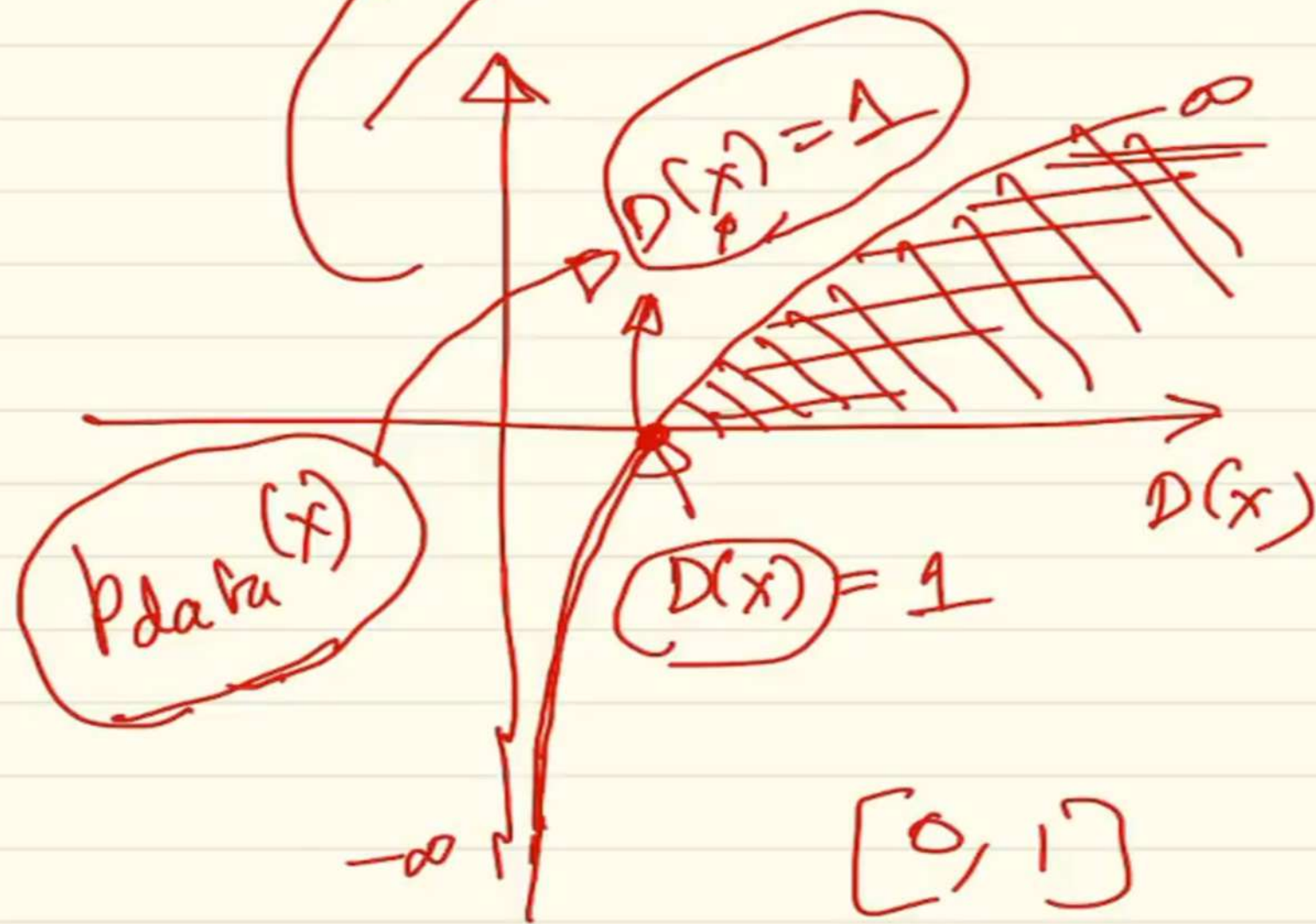
$$L(D(G(x)), 0) = (1-0) \log(1-D(G(x))) = \log(1-D(G(x)))$$

$\textcircled{B}$

Objective of the discriminator is to correctly classify fake
vs the real dataset. For this (A) & (B) should be maximized

(A) $\rightarrow [\log(D(x))]$

$\rightarrow [\log(1 - D(G(z)))]$

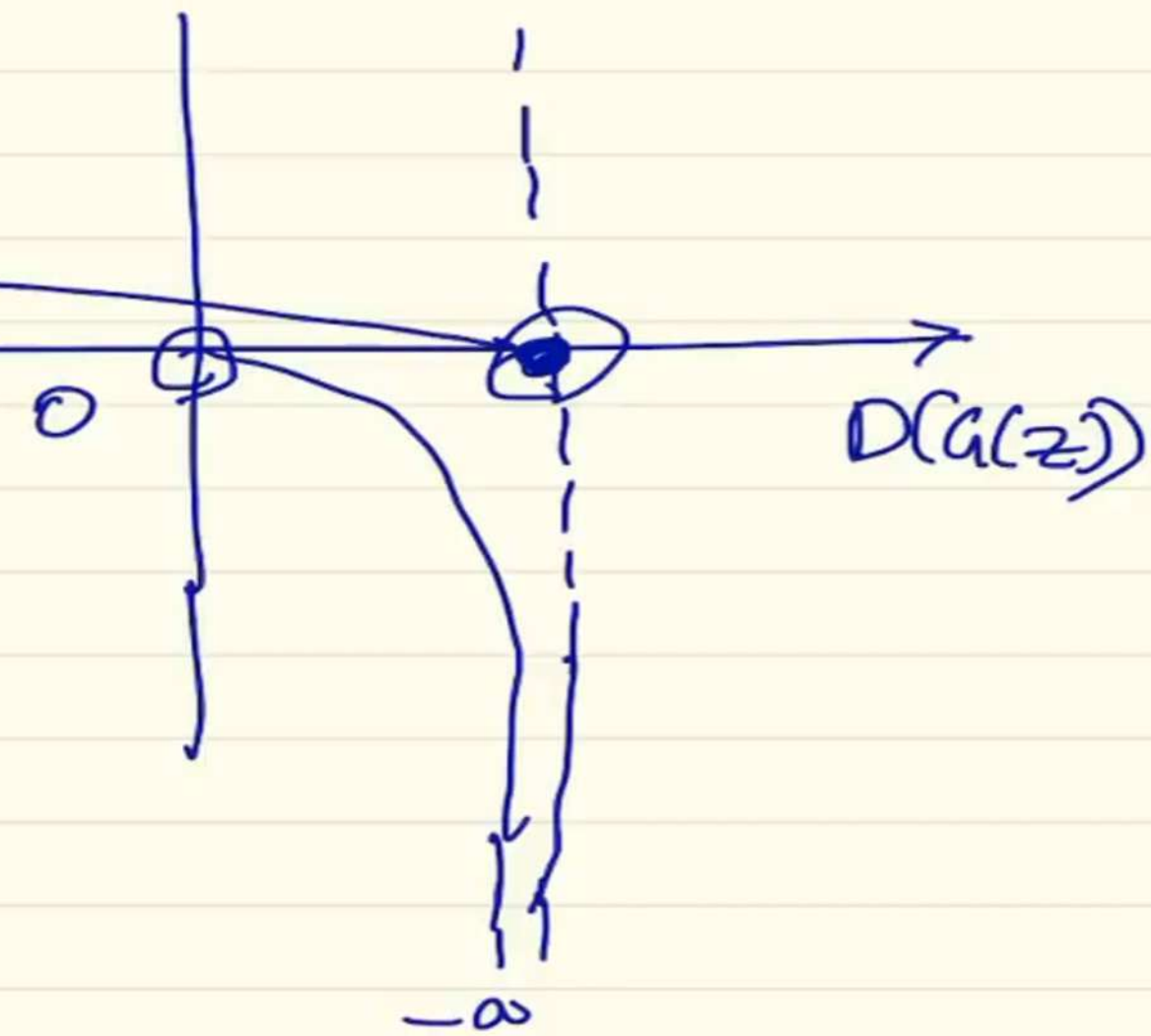


$$\boxed{\max \{ \log(D(x)) + \log(1 - D(G(z))) \}} = \textcircled{D}$$

$$\underline{D(G(z)) = \textcircled{1}} \rightarrow \log(1 - D(G(z)))$$

$$D(G(z)) = \textcircled{1}$$

$$\min [\log(D(x)) + \log(1 - D(G(z)))]$$



$$\min_G \max_D \left\{ \log(D(x)) + \log(1 - D(G(z))) \right\}$$

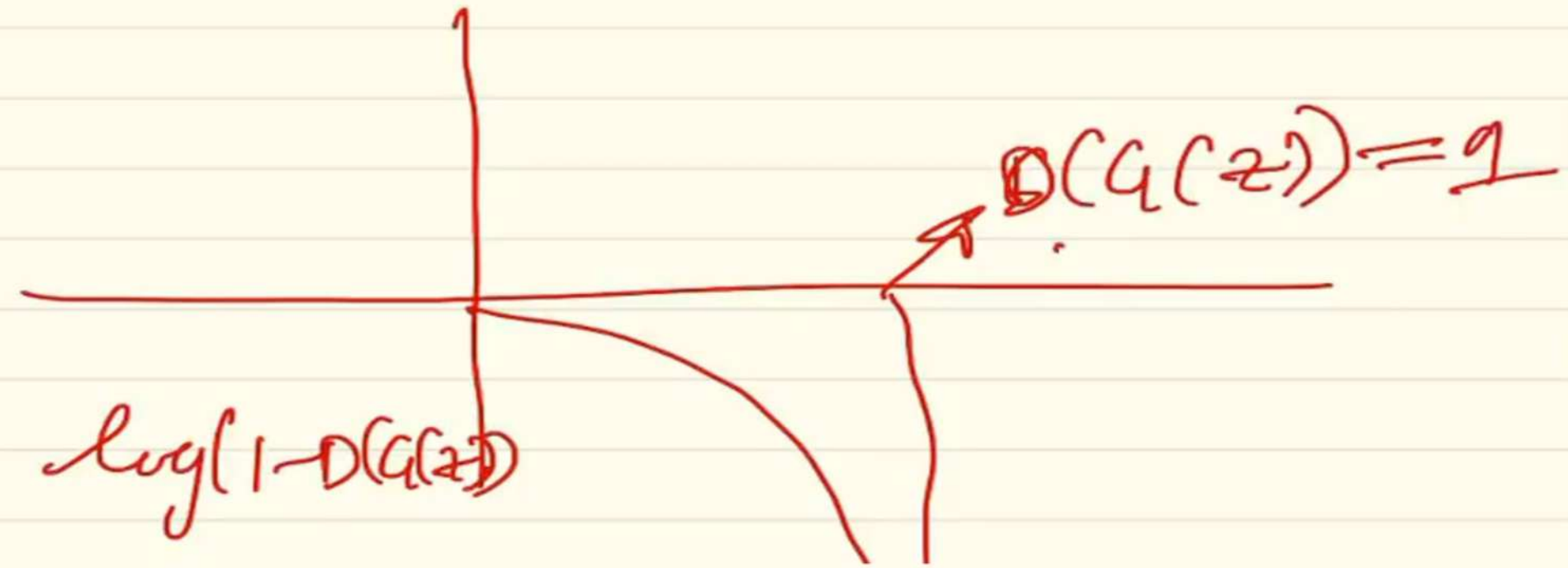
↓ Goodfellow 2014.

$$\min_G \max_D V(D, G) = E_{x \sim p_{\text{data}}(x)} [\log(D(x))] + E_{z \sim p_z} [\log(1 - D(G(z)))]$$

$$\boxed{y=0}, \hat{y} = \boxed{D(G(z))}$$

$$L(\underset{\hat{y}}{D(G(z))}, 0) = \underbrace{(1) \log[1 - D(G(z))]}$$

$$D[G(z)] = 1$$



→ Loss function

$$\min_G \max_D V(D, G) = \min_G \max_D \left\{ E_{x \sim p_{\text{data}}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log (1 - D(G(z)))] \right\} \quad \text{--- (A)}$$

Finding the Best Discriminator

Proposition: For G fixed, the optimal discriminator D is

$$D_G^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)}$$

Proof: The training criterion for the discriminator D , given any generator G is to maximize eq: (A)

Hence, The optimal discriminator for a given G is denoted as $D_G^* = \arg \max_D V(D, G)$

Please note $E_{p(x)}[x] = \int x p_x(x) dx$ — (1)

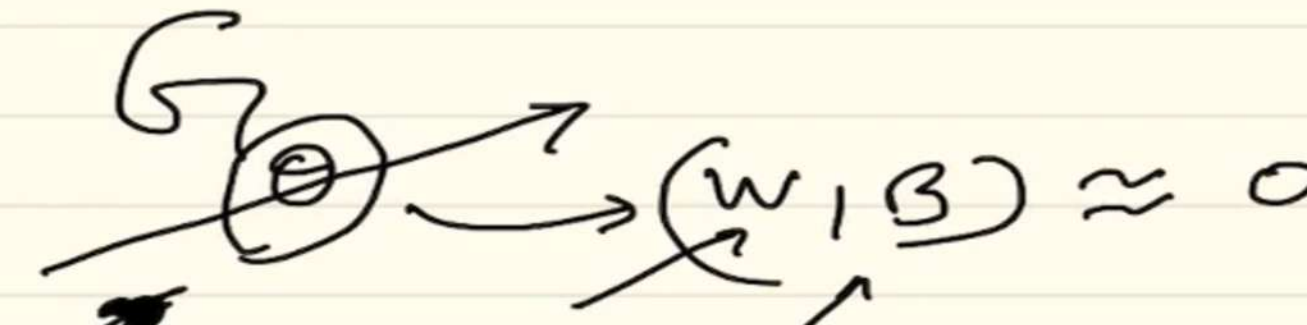
$$D_G^* = \arg \max_D \left\{ E_{x \sim p_{\text{data}}(x)} [\log(D(x))] + E_{z \sim p_z(z)} [\log(1 - D(G(x)))] \right\}$$

$$= \arg \max_D \{ V(G, D) \}$$

(2)

PROBLEMS WITH GAN

① Vanishing Gradients

 $(w, b) \approx 0$ $\frac{\partial}{\partial w}, \frac{\partial}{\partial b} \rightarrow 0$ training process.

① Discriminator $\begin{cases} \text{real} \\ \text{false} \end{cases}$

② Generator $\leftarrow$ noisy



Mode Collapse

epoch 350k



epoch 390k



epoch 396k



During the training, the generator may collapse to a setting where it always produces same output. This is called as **MODE COLLAPSE**

epoch 400k



epoch 480k



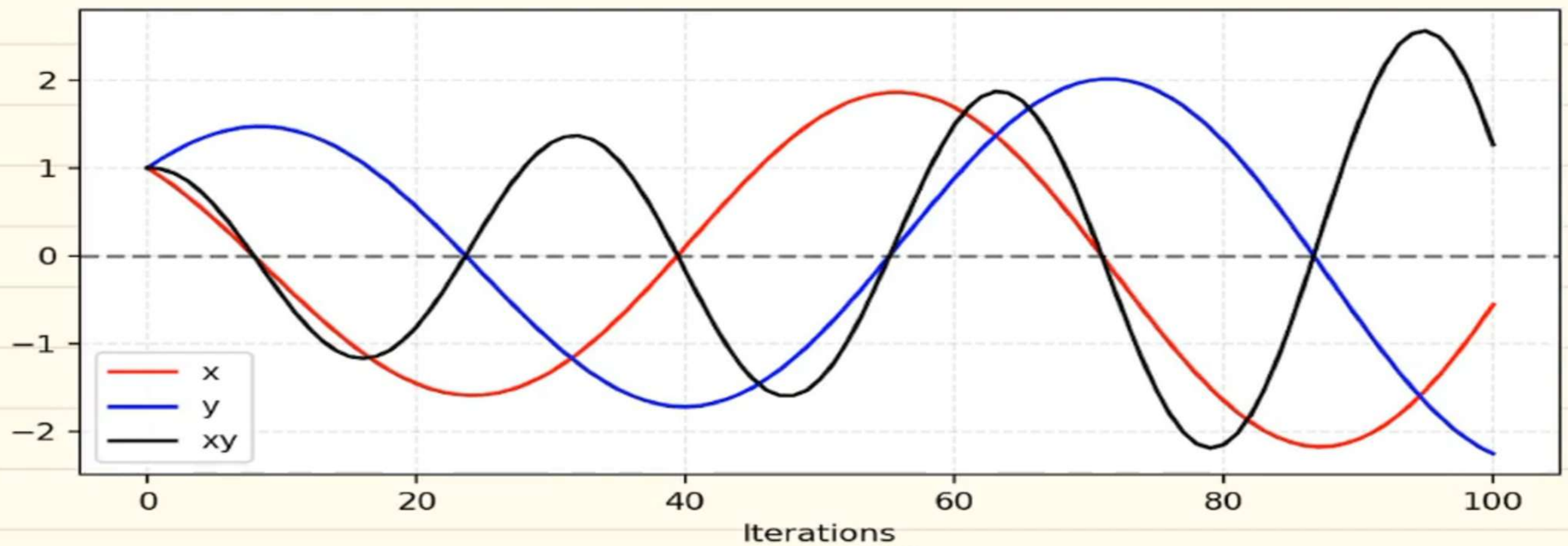
MODE COLLAPSE WITH RESPECT TO MNIST (NUMBER) DATASET



(3) Hard to achieve Nash equilibrium. →

$$f_1(x) = xy \rightarrow \frac{\partial f_1(x)}{\partial x} = y \quad \} \quad x - \eta y$$

$$f_2(y) = -xy \quad \frac{\partial f_2(y)}{\partial y} = -x \quad \} \quad y + \eta x$$



Problems with Counting

✓



(Goodfellow 2016)

Source: NIPS 2016- arXiv: 1701.00160, 2017

CANNOT DISTINGUISH BETWEEN 3D AND 2D OBJECTS



KL DIVERGENCE

Unknow
 P Q

KL Div

(Forward KL)

DL-20

(Reverse KL)

$$KL(P||Q) = \sum p(x) \log \frac{P(x)}{Q(x)}$$

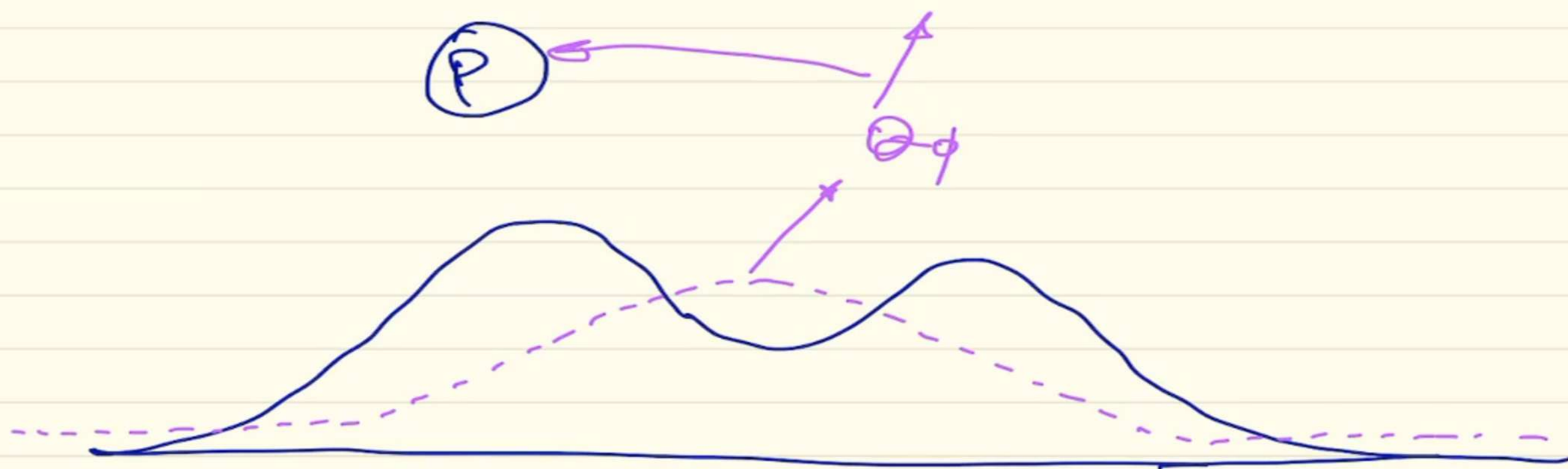
$$KL(Q||P) \equiv \sum Q(x) \log \frac{Q(x)}{P(x)}$$

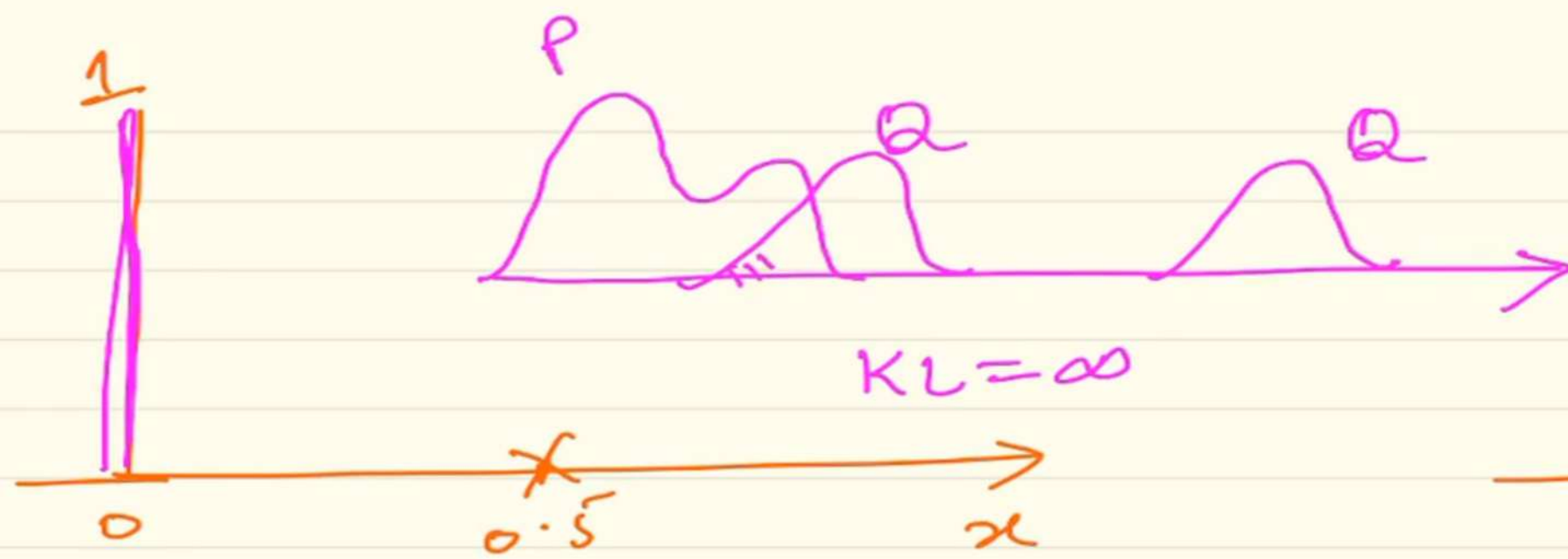
VAE

$$\min (KL(Q||P^\phi))$$

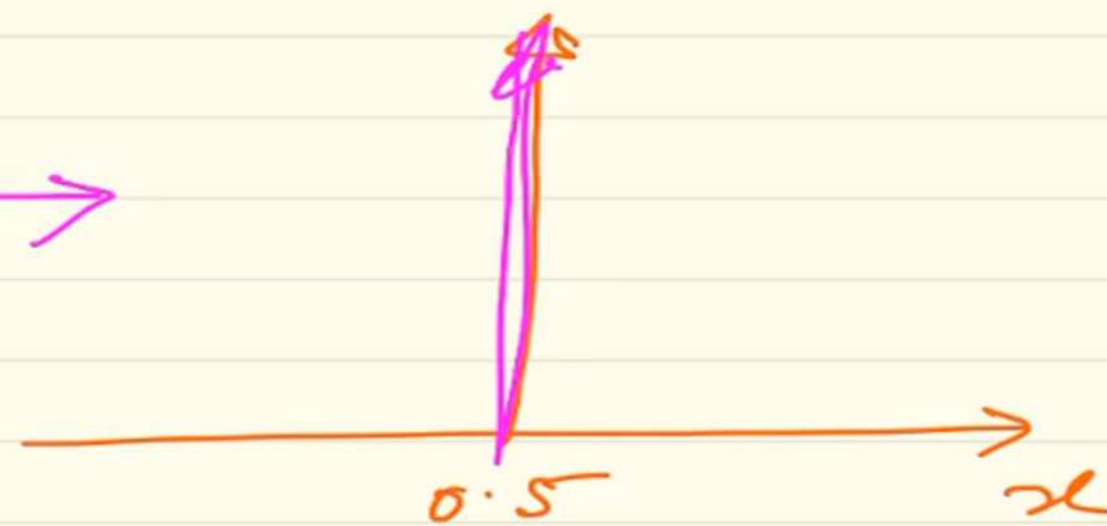
Reverse KL

$$\min_{\phi} \{ \text{KL} \{ Q_{\phi} || P \} \} \Rightarrow \min_{\phi} \left\{ \sum Q_{\phi}^{(x)} \log \frac{Q_{\phi}(x)}{P(x)} \right\}$$





(P)



(Q)

$$KL(P||Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)} = P(x=0) \log \frac{P(x=0)}{Q(x=0)}$$

$$= 1 \log \frac{1}{0} = \infty \checkmark$$

$$KL(Q||P) = \sum_x Q(x) \log \frac{Q(x)}{P(x)} = \overset{1}{Q(x=0.5)} \log \frac{\overset{1}{Q(x=0.5)}}{\underset{0}{P(x=0.5)}} = \infty$$

JANSON SHANON



$$JSD(P||Q) = \frac{1}{2} \left[\underbrace{KL(P||M)} + \underbrace{KL(Q||M)} \right]$$

$$M = \left(\frac{P+Q}{2} \right)$$

$$JSD(P||Q) = \frac{1}{2} \left[\underbrace{P(x=0)}_{\text{"Q"}} \log \frac{P(x=0)}{\underbrace{\frac{P(x=0)+Q(x=0)}{2}}} \right] +$$

KL P

$$= \frac{1}{2} \left[1 \log \frac{1}{1/2} + 1 \log \frac{1}{1/2} \right] = \left[\log 2 \right] \underbrace{2}_{JSD}$$

$$\underbrace{Q(x=0.5)} \log \frac{Q(x=0.5)}{\underbrace{\frac{P(x=0.5)+Q(x=0.5)}{2}}}$$

THANK YOU